

CAI 4104/6108: Machine Learning Engineering

Project Report: **Predictive Insights into Urban Traffic Safety**

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1 Introduction

Motor vehicle collisions pose a critical public safety challenge, often resulting in injuries or fatalities, particularly among pedestrians. These incidents are influenced by a complex interplay of vehicle dynamics, environmental conditions, and human variables. With the escalating number of vehicles and increasing urban density, traffic accidents have intensified as a major public health concern. Recent advancements in data analytics and machine learning have opened new avenues for addressing these issues through enhanced prediction of collision outcomes and the identification of high-risk zones. Our project focuses on utilizing a comprehensive dataset that includes variables such as vehicle types, roadway configurations, weather conditions, and demographic data to predict the likelihood of pedestrian injuries in vehicle collisions and to identify and group areas at higher risk. This approach not only aims to predict occurrences of injuries but also enhances our understanding of spatial risk factors, enabling targeted safety interventions like improved pedestrian crossings and enhanced traffic signal systems [3]. Furthermore, the predictive models developed could be integrated into real-time road safety applications and navigation systems, providing alerts and recommendations that enhance both individual and community safety. By leveraging machine learning techniques, this project aims to significantly reduce pedestrian injuries and improve urban traffic safety, thereby contributing to safer road environments for the people. [5]

2 Approach: Dataset(s) & Pipeline(s)

Dataset(s)

The dataset, sourced from Data.gov, encompasses comprehensive records of motor vehicle collisions spanning several years. It includes details such as collision dates, times, geographical coordinates, vehicle types involved, and various contributing factors. With over 2 million records and 29 features, the dataset provides a robust foundation for statistical analysis and machine learning.

Machine Learning Pipeline

1. Data Preprocessing

The initial stage of the project involves data preprocessing to prepare the data for further analysis and modeling. This includes handling missing values, converting data types, and removing any irrelevant or redundant features. Key preprocessing steps are as follows:

Data Cleaning

- **Handling Missing Values:** Imputation of missing values using mean for continuous variables and mode for categorical variables.
- **Removing Redundant Columns:** Elimination of columns that do not contribute to the predictive analysis to streamline the dataset.

- **Dropping Duplicates:** Removes duplicate rows from the DataFrame to prevent bias toward overrepresented data, enhancing model generalization.
- **Converting Text to Lowercase:** Ensures text data uniformity across the dataset, which is crucial for consistent categorical data processing.
- **Removing Outliers:** Identifies and removes data points that significantly deviate from the norm, promoting model accuracy and robustness.

Feature Engineering

- **New Variables:** Creation of variables such as ‘time of day’ and ‘day of the week’ from existing date and time columns to identify temporal patterns.
- **Categorical to Numeric:** Conversion of categorical variables into dummy variables using one-hot encoding with Pandas’ `get_dummies()` or sklearn’s `LabelEncoder`.
- **Data Type Conversions:** Converts specific columns to appropriate data types, such as converting ‘ZIP CODE’ to numeric values, handling errors by coercing to NaN and replacing them with -1. This ensures that the data types are consistent for modeling.
- **Creation of High-Risk Indicator:** Introduces a binary ‘high_risk’ indicator based on the total count of injuries and fatalities exceeding a set threshold.

Rescaling

- **Standardization:** Application of `StandardScaler` from sklearn, which removes the mean and scales features to unit variance.

2. Model Preparation

- **Train/Test Split:** The dataset is split into training and testing sets using sklearn’s `train_test_split` to ensure the model can be evaluated on unseen data.

3. Model Selection and Training

Various models are explored for best predicting pedestrian injuries:

- **Decision Tree:** Utilization of `DecisionTreeClassifier` from sklearn. Decision Trees are simple to understand and interpret, and they work well when you have categorical input variables. They can, however, be prone to overfitting, especially with complex trees.
- **Random Forest:** Application of `RandomForestClassifier`, known for its robustness against overfitting due to its ensemble approach, where multiple decision trees vote on the final prediction. This model is beneficial for handling high-dimensional data and provides a measure of feature importance, which can be critical for understanding influencing factors.
- **XGBoost:** Use of XGBoost, a powerful gradient boosting framework that is effective in terms of both speed and performance [1]. XGBoost can handle missing data and is capable of doing parallel computations on a single machine, which makes it an excellent choice for large datasets..

4. Model Optimization

- **Hyperparameter Tuning:** Use of Grid Search (`GridSearchCV`) and Random Search to find the best model parameters.
- **Cross-Validation:** Application of different cross-validation techniques to assess the robustness of the models.

5. Performance Evaluation

- **Evaluation Metrics:** Models are evaluated using accuracy score, classification report, and confusion matrix to understand the models’ predictive capabilities.

3 Evaluation Methodology

Prediction of Pedestrian Injuries

- **Precision:** This will measure the accuracy of the predictions in identifying true pedestrian injury cases among all cases labeled as such by the model. High precision indicates that an algorithm returned substantially more relevant results than irrelevant ones, meaning most of the predicted injuries were actual injuries.
- **Recall (Sensitivity):** This will gauge the model’s ability to capture all actual pedestrian injury cases. High recall means the model correctly identified most of the actual injuries.
- **F1 Score:** As a harmonic mean of precision and recall, this score provides a balance between them, which is crucial in scenarios where both false positives and false negatives are significant. The F1 score reaches its best value at 1 (perfect precision and recall) and worst at 0.
- **Accuracy:** This measures the overall correctness of the model across all predictions made.

Predicting High-Risk areas

- **Precision:** In the context of predicting high-risk areas, precision reflects how many of the areas predicted as high-risk actually experienced high frequencies of injuries or fatalities.
- **Recall:** This metric indicates the ability of the model to capture all the actual high-risk areas within the dataset. A model with high recall covers most of the genuine high-risk scenarios.
- **F1 Score:** Similar to the pedestrian injury prediction, the F1 score provides a balanced measure of precision and recall, evaluating the overall efficacy of the model in identifying high-risk areas.
- **Accuracy:** Evaluates the overall performance of the model in correctly identifying risk levels across all areas.

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives}$$

$$Recall = \frac{TruePositives}{TruePositives + FalseNegatives}$$

$$F1 = 2 \cdot \frac{Precision \times Recall}{Precision + Recall}$$

Baselines:

Random Guessing

Description: In our study, random guessing is employed as the baseline model. This approach involves making predictions based solely on the probability distribution of the outcome classes within the dataset. For pedestrian injuries, random guesses are based on the frequency of injury vs. no-injury cases. For the classification of collisions into weekdays or weekends, guesses are aligned with the proportion of weekday and weekend incidents observed in the training data. [2]

Data Split

The dataset is likely split into training and testing sets to effectively validate the models. Typically, a conventional split, such as 80% for training and 20% for testing, might be employed to ensure that the models do not overfit and can generalize well to unseen data.

4 Results

In our project, we compared several machine learning models for predicting pedestrian injuries and identifying high-risk areas. We focused on the performance metrics of accuracy, precision, recall, and F1-score. The results showed that XGBoost outperformed the other models, achieving the highest scores across all evaluated metrics. This superior performance of XGBoost highlights its effectiveness in predictive tasks essential for enhancing urban safety [4]. The successful application of XGBoost in our project demonstrates its potential in predicting pedestrian

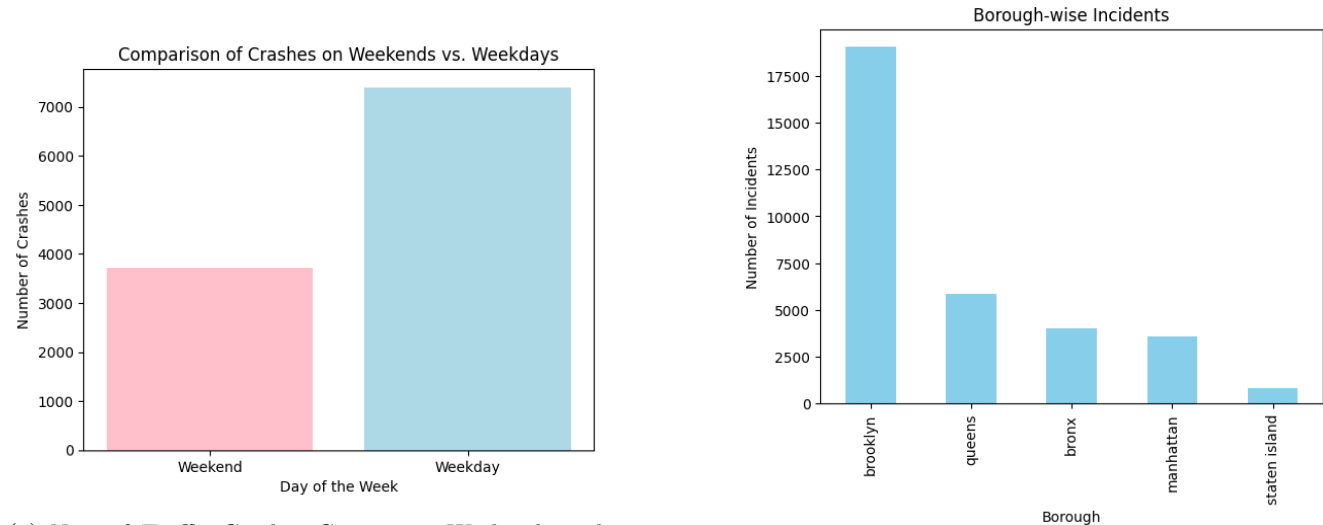
injuries and identifying areas of high risk, which can significantly contribute to targeted interventions and safety improvements.

Model	Metric	Initial	Tuned	Cross-Validation
Decision Tree	Accuracy	65.94%	69.34%	66.30%
	F1 Score	0.66	0.73	0.68
	Precision	0.67	0.73	0.60
	Recall	0.65	0.84	0.75
Random Forest	Accuracy	68.70%	71.01%	70.65%
	F1 Score	0.70	0.75	0.74
	Precision	0.68	0.68	0.66
	Recall	0.72	0.84	0.83
XGBoost	Accuracy	73.05%	73.74%	72.24%
	F1 Score	0.74	0.73	0.77
	Precision	0.66	0.67	0.68
	Recall	0.78	0.80	0.86

Table 1: **Comparative Results of Machine Learning Models for predicting pedestrian injuries**

Model	Metric	Tuned
Decision Tree	Accuracy	56.97%
	F1 Score	0.54
	Precision	0.50
	Recall	0.52
Random Forest	Accuracy	57.11%
	F1 Score	0.58
	Precision	0.55
	Recall	0.57
XGBoost	Accuracy	57.97%
	F1 Score	0.58
	Precision	0.59
	Recall	0.57

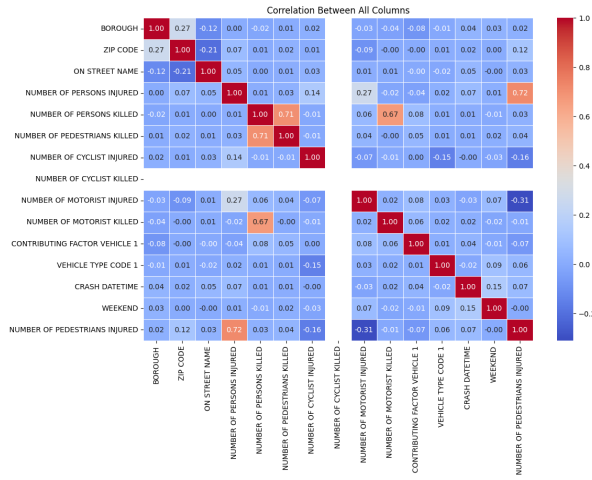
Table 2: **Comparative Results of Machine Learning Models for predicting high risk areas**



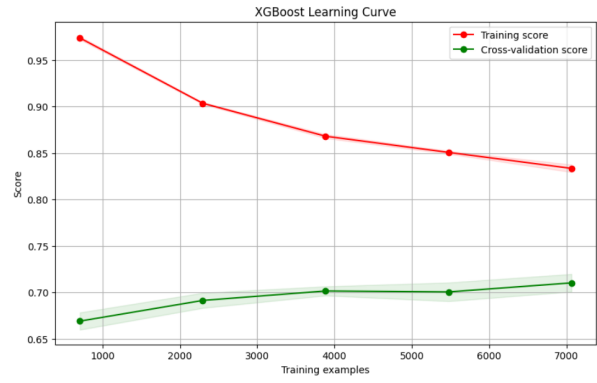
(a) No. of Traffic Crashes Comparing Weekends and Weekdays

(b) Borough wise incidents

Figure 1: In figure a, Bar Chart Showing the Number of Traffic Crashes Comparing Weekends and Weekdays: A clear increase in the number of incidents is observed on weekdays, suggesting a higher volume of vehicular activity during these days. In figure b, Illustrates the number of reported incidents in each of the five boroughs of New York City, with Brooklyn showing the highest incidence and Staten Island the least.



(a) Correlation between all columns



(b) Learning curve for predicting pedestrian injuries

Figure 2: In figure a, Correlation heatmap showing the relationships between various attributes in a dataset, with values ranging from -1 to 1 indicating the strength and direction of the correlation. In figure b, A graph depicting the learning curve of an XGBoost model used to predict pedestrian injuries, showing the relationship between training size and model performance for both training and cross-validation scores.

5 Conclusions

The project has successfully demonstrated how machine learning can be applied to enhance urban traffic safety by predicting pedestrian injuries and identifying high-risk areas. Utilizing a comprehensive dataset and sophisticated models, particularly XGBoost, the project achieved impressive predictive performance, underscoring the potential of technology to address significant public safety challenges. This success paves the way for integrating these insights into real-time safety applications and urban planning tools, potentially reducing pedestrian injuries. Looking forward, improvements could include enriching the dataset with real-time traffic and weather data, experimenting with various machine learning algorithms, and developing real-time applications for immediate alerts. Additionally, applying the model to diverse urban environments and assessing the real-world impact of the predictions could further refine its effectiveness and scalability. Automating feature engineering and optimizing model configurations through AutoML could also enhance the model's efficiency and performance, solidifying its utility in promoting safer urban environments. The exploration of additional models such as SVM, GBM, AdaBoost, LSTM networks, transfer learning, autoencoders, and Bayesian methods could potentially uncover new insights and improve the predictive performance of the models in urban traffic safety.

References

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